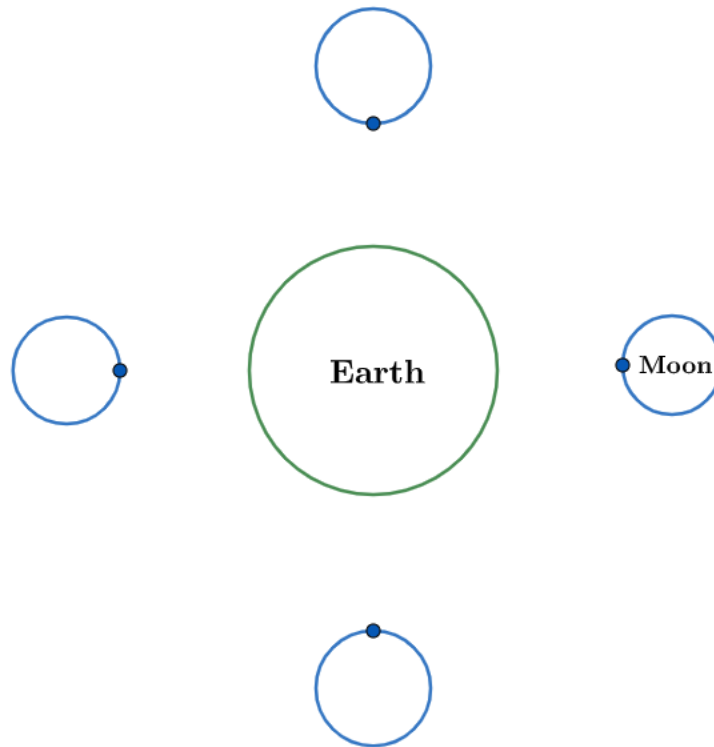


2021 F=ma Exam: Problem 20

Kevin S. Huang



Note the speed of the Moon's terminator is the tangential speed of the Moon's surface from the rotation about its axis:

$$v = R\omega = R \left(\frac{2\pi}{T} \right)$$

Using the fact that one side of the Moon always faces the Earth, the period of the Moon's rotation is the same as the period of the Moon's revolution around the Earth (1 month). Thus,

$$v = \frac{2\pi(1.74 \times 10^6 \text{ m})}{1 \text{ month}} = 4.5 \text{ m/s}$$

so the answer is B.